

Fig. 1.14 Integrable dynamics may be viewed as periodic or quasi-periodic motion on the surface of a torus (illustrated for a system with two degrees of freedom)

1.5.3 Equilibrium Points, and the Linearization Theorem

Molecular systems can be written in the form

$$\frac{\mathrm{d}z}{\mathrm{d}t} = \boldsymbol{f}(z). \tag{1.6}$$

An *equilibrium point* of such a system is a solution of $\boldsymbol{f}(z) = 0$.[4] An equilibrium point z^* corresponds to an equilibrium solution, since if we define a constant function $z(t) \equiv z^*$ then we have $\dot{z}(t) \equiv \boldsymbol{f}(z^*) = 0$. In some cases, it is possible to discuss the motion of solutions close to equilibrium points by means of *linearization*: the technique of replacing a nonlinear system by a linear one which approximates it locally.

We assume that $\boldsymbol{f}$ is continuously differentiable in the vicinity of the equilibrium point z^* and make use of the fact that $\boldsymbol{f}(z) \approx \boldsymbol{f}(z^*) + \boldsymbol{f}'(z^*)(z - z^*)$ for $\|z - z^*\|$ sufficiently small. Since $\boldsymbol{f}(z^*) = 0$ we have, defining $\delta z := z - z^*$,

$$\frac{\mathrm{d}\delta z}{\mathrm{d}t} \approx \boldsymbol{A}\delta z, \qquad \boldsymbol{A} = \boldsymbol{f}'(z^*).$$

The symbol $\approx$ is not very precise. In general, the approximation of solutions is only local in time meaning that solutions of the nonlinear system and the linear approximation may diverge. We refer to the system

$$\frac{\mathrm{d}\delta z}{\mathrm{d}t} = \boldsymbol{f}'(z^*)\delta z$$

as the *linearization* of the system (1.6) around z^*.

In case the equilibrium point is *hyperbolic*, meaning that the real parts of the eigenvalues of $\boldsymbol{A} = \boldsymbol{f}'(z^*)$ are nonzero, then one can infer that the solutions of the

[4]For readers conversant with concepts of statistical mechanics and in preparation for later discussions, we emphasize that the concept of *mechanical equilibrium* given in relation to the system of ordinary differential equation (1.6) is entirely different from the notion of *statistical* or *thermal equilibrium* of Chap. 6.

nonlinear and linear systems are in fact *topologically conjugate*: if z is the solution of the nonlinear system and δz is the solution of the linear system, then there is a smooth, invertible map $\boldsymbol{\Phi}$ of $\mathbb{R}^m$ defined in a neighborhood of the origin such that

$$z(t) = z^* + \boldsymbol{\Phi}(\delta z(t)).$$

This is referred to as the Hartman-Grobman theorem (for more discussion, see [177], where this result is referred to as the "Linearization Theorem"; a proof may be found in [362]).

Let z^* be an equilibrium point. We say that z^* is stable ("in the sense of Lyapunov") if, for all ε, there exists δ such that, for all z_0 such that $\|z_0 - z^*\| < \delta$,

$$\sup_{t \geq 0} \|\mathscr{F}_t(z_0) - z^*\| < \varepsilon.$$

That is, the solution started near to z^* stays near z^* for all time. The Hartman-Grobman theorem clearly implies that the stability of a given hyperbolic equilibrium point z^* of a nonlinear system can be inferred from the stability of the origin for the linearization of the system around z^*.

In the case of Hamiltonian systems, the eigenvalues are, very often, on the imaginary axis, and the Hartman-Grobman theorem is not of much use. However, there is another result which is relevant in this case. Observe that an equilibrium point $z^* = (\boldsymbol{q}^*, \boldsymbol{p}^*)$ of a Hamiltonian system in "kinetic plus potential" form

$$H(\boldsymbol{q}, \boldsymbol{p}) = \boldsymbol{p}^T \boldsymbol{M}^{-1} \boldsymbol{p}/2 + U(\boldsymbol{q})$$

will always have $\boldsymbol{p}^* = 0$ and $\nabla U(\boldsymbol{q}^*) = 0$. We say that $\boldsymbol{q}^*$ is a *strong local minimum* of the potential if there exists $\varepsilon > 0$ such that

$$0 < \|\boldsymbol{q} - \boldsymbol{q}^*\| < \varepsilon \Rightarrow U(\boldsymbol{q}) > U(\boldsymbol{q}^*).$$

Then we have

Theorem 1.1 *If $\boldsymbol{q}^*$ is a strong local minimum of smooth potential U then $z^* = (\boldsymbol{q}^*, \boldsymbol{0})$ is stable.*

The proof of this theorem relies on showing that if trajectories are started from a point sufficiently close to z^* they cannot wander away to infinity. Although the result holds in greater generality, it is easy to show under assumptions of local smoothness of U (which we are normally happy to make in molecular dynamics). For more discussion see the text [216]. If the potential is C^2, the linearized version of this system has "kinetic plus potential" form with Hamiltonian

$$\bar{H} = \frac{\delta \boldsymbol{p}^T \boldsymbol{M}^{-1} \delta \boldsymbol{p}}{2} + \frac{\delta \boldsymbol{q}^T U''(\boldsymbol{q}^*) \delta \boldsymbol{q}}{2}.$$